

Notes: Liu (QJE, 2019)

Industrial Policies in Production Networks

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1 Big picture

- The paper asks: *when do sectoral industrial policies raise aggregate output / welfare in a production network?*
- Key idea: in networks, distortions “compound” along supply chains. Some sectors are *too small relative to how important they would be in an efficient allocation*. Those sectors are the ones where a marginal subsidy has the biggest social value.
- The main sufficient statistic is **distortion centrality** ξ_i , which is (roughly) “potential size” / “actual size” of sector i .
- Empirical punchline: Input–output (I–O) demand matrices for South Korea (1970) and China (2007) are close to *hierarchical* (upstream-to-downstream). In such networks, ξ_i is very stable across different ways of modeling / estimating frictions. In practice, ξ_i is highly correlated with standard upstreamness measures.
- Applications:
 1. South Korea’s Heavy and Chemical Industry (HCI) drive in the 1970s: targeted sectors are strongly upstream / high- ξ .
 2. Modern China: sectors with high ξ receive more favorable credit and tax treatment, and have larger SOE presence; implied output gains from these interventions are positive and sizable.

2 Model (static production network with wedges + subsidies)

There are N intermediate sectors $i = 1, \dots, N$ and a “consumption good” produced from intermediates.

2.1 Technologies

Final (gross) output:

$$Y^G = F(Y_1, \dots, Y_N), \tag{1}$$

where Y_i is the amount of intermediate good i used in final-good production.

Intermediate sector i :

$$Q_i = z_i F_i\left(L_i, \{M_{ij}\}_{j=1}^N\right), \tag{2}$$

where L_i is labor and M_{ij} is intermediate input j used by sector i .

2.2 Wedges and policy interventions

- **Market imperfections (wedges):** when sector i purchases input j , it pays an extra proportional cost $\chi_{ij} \geq 0$ per dollar of input expenditure. (The paper interprets these as deadweight-loss generating frictions / “quasi-rents” rather than iceberg transport costs.)

- **Industrial policies:** the government can subsidize intermediate purchases τ_{ij} and/or labor (value-added) input τ_{iL} .

Unit-cost minimization for good i :

$$P_i \equiv \min_{\ell_i, \{m_{ij}\}} \left\{ \sum_{j=1}^N (1 - \tau_{ij} + \chi_{ij}) P_j m_{ij} + (1 - \tau_{iL}) W \ell_i \right\} \quad \text{s.t.} \quad z_i F_i(\ell_i, \{m_{ij}\}) \geq 1. \quad (3)$$

Here (ℓ_i, m_{ij}) are *unit* input requirements, W is the wage, and P_i is the equilibrium unit cost (price).

2.3 Accounting: deadweight loss, budgets, market clearing

Total deadweight loss from wedges:

$$\Pi \equiv \sum_{i=1}^N \sum_{j=1}^N \chi_{ij} P_j M_{ij}. \quad (4)$$

Final-good price normalization (consumption good is numeraire):

$$1 = \min_{\{Y_j\}} \sum_{j=1}^N P_j Y_j \quad \text{s.t.} \quad F(Y_1, \dots, Y_N) = 1. \quad (5)$$

Policy expenditure (subsidy bill) in sector i :

$$S_i \equiv \sum_{j=1}^N \tau_{ij} P_j M_{ij} + \tau_{iL} W L_i. \quad (6)$$

Government budget:

$$G + \sum_{i=1}^N S_i = T, \quad (7)$$

where G is public consumption and T is a lump-sum tax on the representative household.

Household budget:

$$C = WL - T, \quad (8)$$

with total labor supply L (exogenous).

Net output identity:

$$Y^G - \Pi = Y = C + G. \quad (9)$$

Income accounting:

$$Y = WL - \sum_{i=1}^N S_i. \quad (10)$$

Market clearing:

$$L = \sum_{i=1}^N L_i, \quad (11)$$

$$Q_j = Y_j + \sum_{i=1}^N M_{ij} \quad \forall j. \quad (12)$$

3 Network objects and centralities

The paper uses two key “centrality” concepts: **influence** and **Domar weights**. Their ratio is **distortion centrality**.

3.1 Elasticity matrix Σ and expenditure share matrix Ω

Define the (equilibrium) production elasticities

$$\sigma_{ij} \equiv \frac{\partial \ln F_i(L_i, \{M_{ik}\})}{\partial \ln M_{ij}}, \quad \sigma_{iL} \equiv \frac{\partial \ln F_i(L_i, \{M_{ik}\})}{\partial \ln L_i}. \quad (13)$$

Collect σ_{ij} into $\Sigma \in \mathbb{R}^{N \times N}$ and σ_{iL} into $\sigma_L \in \mathbb{R}^N$.

Define (equilibrium) expenditure shares

$$\omega_{ij} \equiv \frac{P_j M_{ij}}{P_i Q_i}, \quad \omega_{iL} \equiv \frac{W L_i}{P_i Q_i}, \quad (14)$$

and collect into $\Omega \in \mathbb{R}^{N \times N}$ and $\omega_L \in \mathbb{R}^N$.

Because wedges/subsidies distort the link between elasticities and spending shares,

$$(1 + \chi_{ij} - \tau_{ij}) \omega_{ij} = \sigma_{ij}, \quad (15)$$

$$(1 - \tau_{iL}) \omega_{iL} = \sigma_{iL}. \quad (16)$$

3.2 Final-demand shares and *influence* μ

Let β_i be the expenditure share of intermediate good i in producing the final good:

$$\beta_i \equiv \frac{P_i Y_i}{\sum_{k=1}^N P_k Y_k}. \quad (17)$$

Influence is an elasticity-based centrality:

$$\mu' \equiv \beta'(I - \Sigma)^{-1}. \quad (18)$$

Interpretation: μ_i is the elasticity of aggregate factor income WL w.r.t. sector i productivity:

$$\frac{d \ln(WL)}{d \ln z_i} = \mu_i. \quad (19)$$

So μ_i captures how a local productivity change propagates through the network.

3.3 Domar weights γ

Domar weight (size centrality) of sector i :

$$\gamma_i \equiv \frac{P_i Q_i}{WL}. \quad (20)$$

Interpretation: value of gross output in sector i relative to total factor income.

3.4 Distortion centrality ξ

$$\xi_i \equiv \frac{\mu_i}{\gamma_i}. \quad (21)$$

Interpretation (the way we internalize it):

- μ_i is “how big sector i *wants* to be” (importance in an efficient propagation sense).
- γ_i is “how big sector i *is*” in the distorted equilibrium.
- $\xi_i > 1$ means sector i is *too small* relative to its network importance; marginally expanding it has high social value.
- In an efficient (undistorted) economy, $\xi_i = 1$ for all i .

4 Main theoretical results: welfare effects of industrial policy

All the key welfare statements are *first-order* (local) statements around the **decentralized economy** with wedges χ_{ij} but *no subsidies* ($\tau = 0$).

4.1 Lemma: marginal output effect of a subsidy

For an intermediate-input subsidy τ_{ij} (or labor subsidy with $j = L$),

$$\left. \frac{d \ln Y}{d \tau_{ij}} \right|_{\tau=0} = \omega_{ij} (\mu_i - \gamma_i) = \omega_{ij} \gamma_i (\xi_i - 1). \quad (22)$$

So the sign is governed by $(\xi_i - 1)$: subsidizing sector i raises aggregate output iff $\xi_i > 1$.

4.2 Proposition 1: *social value* of policy expenditure is ξ_i

Holding the lump-sum tax T fixed, a subsidy changes private and public consumption. Define the social value of a subsidy on input j in sector i as

$$SV_{ij} \equiv \left. \frac{dC/d\tau_{ij}}{dG/d\tau_{ij}} \right|_{\tau=0, T \text{ fixed}}.$$

Then

$$SV_{ij} = \xi_i \quad \text{for all } j \in \{1, \dots, N, L\}. \quad (23)$$

Two things I find striking:

- It does *not* matter whether the subsidy is on labor or on some intermediate input: the social value depends only on the sector i being targeted.
- ξ_i is a *sufficient statistic* for the marginal cost–benefit of spending in sector i .

4.3 Corollary: output gain per dollar of subsidy is $\xi_i - 1$

If instead I hold G fixed (and finance the subsidy via lump-sum taxes), then

$$\left. \frac{dY/d\tau_{ij}}{dT/d\tau_{ij}} \right|_{\tau=0, G \text{ fixed}} = \xi_i - 1. \quad (24)$$

4.4 Proposition 2: aggregate gains depend on covariance between spending and ξ

Let sector- i policy spending relative to factor payments be

$$s_i \equiv \frac{S_i}{WL_i}.$$

Then, for small interventions, proportional output gains are approximately

$$\Delta \ln Y \approx \text{Cov}(\xi_i, s_i), \quad (25)$$

where the cross-sector covariance is taken using sectoral labor shares as weights.

Interpretation: industrial policy works (to first order) when it *targets* high- ξ sectors, i.e. when s_i is positively correlated with ξ_i .

4.5 Proposition 3 (Cobb–Douglas): optimal value-added subsidies

In the Cobb–Douglas case, if the government can choose only sectoral labor/value-added subsidies $\{\tau_{iL}\}$, then the output-maximizing subsidies satisfy

$$\frac{1}{1 - \tau_{iL}^*} = \xi_i. \quad (26)$$

This aligns distorted allocations with the efficient allocation:

$$L_i^* = \mu_i \sigma_{iL} L \quad (\text{efficient}), \quad (27)$$

$$L_i = \frac{\gamma_i}{1 - \tau_{iL}} \sigma_{iL} L \quad (\text{with value-added subsidies}). \quad (28)$$

So setting $\gamma_i/(1 - \tau_{iL}) = \mu_i$ exactly corrects the sectoral scale distortion.

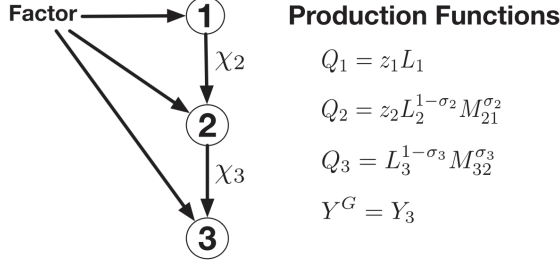


FIGURE I

A Vertical Production Network with Three Intermediate Sectors

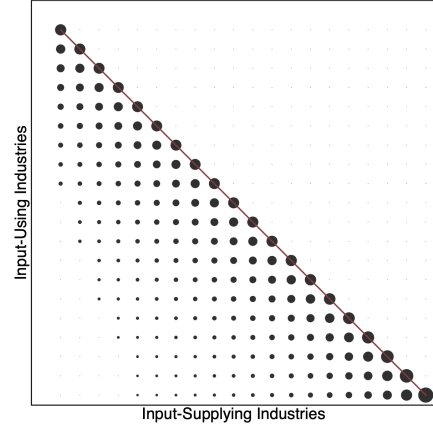


FIGURE II

An Illustrative Input-Output Demand Matrix of a Hierarchical Network

5 Network structure and why “upstream” sectors often have high ξ

5.1 I–O demand matrix Θ

Define

$$\theta_{ij} \equiv \frac{M_{ij}}{Q_j}, \quad (29)$$

the fraction of sector j ’s gross output sold to sector i (“who buys j ’s output?”).

Let $\theta_j^F \equiv Y_j/Q_j$ be the share of good j that goes to final consumption. Market clearing gives $\theta_j^F + \sum_{i=1}^N \theta_{ij} = 1$.

Collect θ_{ij} into $\Theta \in \mathbb{R}^{N \times N}$ and θ^F into a vector.

Relationship to expenditure shares:

$$\theta_{ij} = \omega_{ij} \frac{\gamma_i}{\gamma_j}. \quad (30)$$

5.2 Proposition 4: a recursion for ξ (“distortions propagate upstream”)

Let $D = [\chi_{ij}]$ be the matrix of wedges and let $\delta \equiv \frac{WL}{Y\bar{G}}$. Then in the decentralized economy,

$$\xi_j = \delta \theta_j^F + \sum_{i=1}^N \xi_i (1 + \chi_{ij}) \theta_{ij}. \quad (31)$$

In matrix form,

$$\xi' = \delta (\theta^F)' \left(I - (\Theta + D \circ \Theta) \right)^{-1}, \quad (32)$$

where $\circ$ denotes the Hadamard (element-by-element) product.

Intuition: if sector j mainly sells to buyers i that themselves have high ξ_i , and if selling to those buyers is associated with large wedges $(1 + \chi_{ij})$, then sector j inherits high distortion centrality. Because selling “upstream” means selling into the whole rest of the network, frictions get multiplied, and ξ tends to rise upstream.

5.3 Connection to upstreamness

The (Antràs et al.) upstreamness measure is

$$U' = \mathbf{1}'(I - \Theta)^{-1}. \quad (33)$$

When wedges are small / the network is close to hierarchical, U and ξ move together strongly (this is what the empirical section shows).

5.4 Hierarchical networks (Definition and ranking result)

A production network is **hierarchical** if, after ordering sectors appropriately,

$$\sum_{k=1}^K \theta_{ki} \geq \sum_{k=1}^K \theta_{kj} \quad \forall i < j, \forall K \leq N. \quad (34)$$

Interpretation: higher-ranked (more upstream) sectors sell relatively more to other upstream sectors; the I–O matrix becomes “mostly lower triangular.”

Key result (Proposition 5, informal):

- If the network is hierarchical (and wedges are not too perverse), then ξ_i is *monotone in upstreamness* (upstream sectors have higher ξ).
- Even if wedges are random i.i.d. across input pairs, the *expected* ranking of ξ is monotone in upstreamness in a lower-triangular network.

So, in practice, upstreamness is a useful proxy for “where distortions compound.”

6 Empirics: how the paper recovers (and stress-tests) ξ

6.1 Step 1: show the I–O matrices are approximately hierarchical

Figure III (I–O plots) and Table I (partial-column-sum tests) show that both South Korea (1970) and China (2007) I–O demand matrices satisfy the hierarchical property for the vast majority of comparisons.

6.2 Step 2: show ξ is robust to how wedges are modeled

Table II is the key robustness table:

- Panel A: simulate χ_{ij} from many distributions (different means/variances/tails). ξ remains extremely correlated with the benchmark.
- Panel B: estimate wedges using several strategies (markups/production elasticities, “foreign firms face fewer frictions,” external finance dependence, profit-share proxies). Again, ξ is highly correlated with the benchmark.
- Upstreamness is also very highly correlated with ξ .

6.3 Open economy adjustment

To account for imports/exports, the paper adds a “trade intermediary” sector that converts exports into imports (constant returns). Table III shows the open-economy adjustment barely changes ξ in either country.

7 South Korea 1970s: the Heavy and Chemical Industry (HCI) drive

7.1 What we take away

- The HCI drive picked sectors that were very upstream / high ξ (Figure IV, Table VII).
- Using Proposition 2, reallocating the same total subsidy bill toward the highest- ξ sectors would have generated even larger output gains (Table VIII).

7.2 Figure IV

The figure overlays HCI sectors on the I–O demand matrix. HCI sectors appear upstream (near the “input suppliers” side): they sell a lot to the rest of the network, and do not heavily rely on downstream inputs.

7.3 Table VII (HCI vs non-HCI)

Across essentially every ξ specification, HCI sectors have:

- higher mean ξ than non-HCI sectors,
- and a much higher probability of having $\xi_i > 1$ (often close to 100% for HCI sectors).

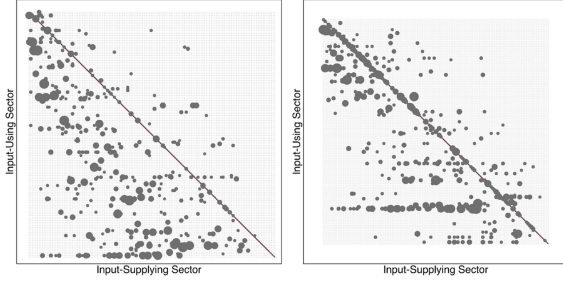


FIGURE III
The IO Demand Matrices of South Korea (Left) and China (Right) Are Hierarchical

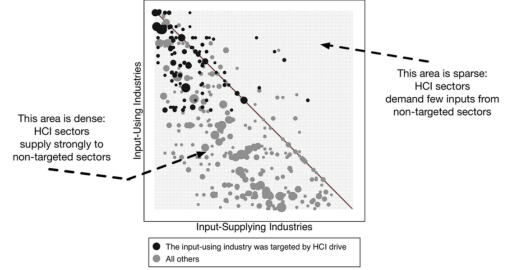


FIGURE IV
South Korea's IO Demand Matrix, with HCI Sectors Darkened

7.4 Table VIII (counterfactual targeting in Korea)

Holding fixed the number of promoted sectors and the total spending:

- Targeting by Domar weights or consumption shares performs poorly (often negative or small gains), consistent with the idea that “big” or “consumer-facing” sectors are not where distortions compound.
- Targeting by export intensity or intermediate-input intensity does better (they correlate with upstreamness).
- Targeting directly by ξ performs best by construction.

TABLE VII
HCI SECTORS HAVE HIGHER DISTORTION CENTRALITY THAN NONTARGETED ONES

ξ Specification	$sd(\xi)$	ξ_i of HCI sectors	Share of sectors with $\xi_i > 1$	
			HCI	Non-HCI
Benchmark	0.09	1.16	100%	47.8%
B3 Rajan and Zingales	0.06	1.12	100%	47.0%
B4 Sectoral profit share	0.16	1.28	100%	45.1%
A3 $N(0.1, 0.1)$	0.09	1.17	100%	47.7%
A7 $U[0, 0.2]$	0.09	1.16	100%	47.7%
A8 $Exp(0.1)$	0.10	1.17	100%	47.7%

Notes. For the simulated specifications, the reported distortion centrality is averaged across 10,000 simulations draws.

TABLE VIII
POLICY COUNTERFACTUALS FOR SOUTH KOREA

Specification for ξ_i :	Average distortion centrality			Gains relative to HCI drive		
	Benchmark (1)	B3 (2)	B4 (3)	Benchmark (4)	B3 (5)	B4 (6)
HCI Drive	1.16	1.12	1.28	100%	100%	100%
Counterfactuals (select sector sorted by...)						
CF1 Domar weight γ	0.98	0.99	0.96	-11%	-9%	-13%
CF2 Consumption share β	0.97	0.94	0.94	-18%	-16%	-22%
CF3 Export intensity	1.07	1.05	1.11	46%	44%	40%
CF4 Sectoral value added	0.98	0.99	0.98	-10%	-9%	-8%
CF5 Intern. exp. share	1.07	1.04	1.08	41%	36%	28%
CF6 Distortion centrality ξ	1.22	1.15	1.30	137%	124%	109%
CF7 Uniform promotion	1	1	1	0%	0%	0%

8 Modern China: who gets favorable policies, and how much does it matter?

8.1 Policy outcomes studied

Using firm-level and sector-level data, the paper studies (mostly within manufacturing sectors):

- **Credit:** effective interest rates and leverage (debt/assets).
- **Taxes:** probability of tax incentives and effective tax rates.
- **Subsidies:** whether firms receive subsidies and subsidy intensity.
- **State ownership:** SOE presence in sectoral value added (and entry of new SOEs).

8.2 Reduced-form sector regressions

A key empirical model is:

$$\text{PolicyOutcome}_i = \alpha + b\xi_i^{10\%} + \Gamma' X_i + \varepsilon_i, \quad (35)$$

where X_i are sector controls (capital intensity, market power/Lerner index, export intensity, etc.).

8.3 Tables IX–X: descriptive stats

These tables summarize the cross-sector distribution of the policy variables (and how they differ between private firms and SOEs). The patterns are consistent with:

- SOEs having cheaper credit and more leverage,
- SOEs receiving more subsidies and paying lower effective taxes on average.

TABLE IX
DESCRIPTIVE STATISTICS: SECTORAL POLICIES VARY SIGNIFICANTLY ACROSS CHINESE SECTORS

Sectoral means (in percentage points)	Min	1st quartile	Median	3rd quartile	Max	Average	Std. dev
Effective interest rate	1.85	3.39	4.12	5.00	12.33	4.45	1.67
Debt ratio	40.91	51.54	54.78	57.04	65.39	54.45	4.82
Fraction of firms with tax incentives	8.70	24.88	30.81	36.55	61.33	31.23	9.84
Effective corporate income tax rate	9.08	15.13	17.48	19.45	24.78	17.29	2.94
Fraction of firms receiving subsidies	6.65	9.62	11.77	13.93	28.83	12.42	3.83
Subsidies/revenue	0.60	0.98	1.36	1.79	4.74	1.57	0.83
SOE share of sectoral value-added	0.66	4.76	10.67	24.40	74.50	17.32	17.04

TABLE X
DESCRIPTIVE STATISTICS: SOES RECEIVE MORE FAVORABLE POLICIES

Variable means by ownership	Private firms	SOEs
Effective interest rate	4.63	2.23
Debt ratio	54.38	63.49
Fraction of firms with tax incentives	31.93	31.48
Effective corporate income tax rate	17.29	14.98
Fraction of firms receiving subsidies	11.46	22.41
Subsidies/revenue	1.56	2.78

8.4 Table XI: policy outcomes for private firms vs ξ

Main message: private firms in high- ξ sectors receive more favorable *credit* and *tax* treatment. Qualitatively:

- higher $\xi \Rightarrow$ lower effective interest rates,
- higher $\xi \Rightarrow$ higher debt ratios,
- higher $\xi \Rightarrow$ more likely to get tax incentives and lower effective tax rates,
- subsidies (as measured) do not load as strongly on ξ .

8.5 Table XII: SOEs and ξ

High- ξ sectors have higher SOE presence. Moreover, high- ξ sectors are also where *new* SOEs are disproportionately established (using different “SOE established after year t ” definitions).

8.6 Table XIII: implied output gains from China’s interventions

Using Proposition 2 and mapping each intervention into sectoral spending s_i :

- credit subsidies and tax incentives are positively correlated with ξ and contribute positive output gains;

TABLE XI
PRIVATE FIRMS IN CHINESE MANUFACTURING SECTORS WITH HIGH DISTORTION CENTRALITY RECEIVE MORE-FAVORABLE POLICIES

	Effective interest rate	Debt ratio		Tax break		Effective tax rate	Recipient of subsidies		Subsidies received
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$\xi^{10\%}$	-0.895*** (0.222)	-0.987*** (0.223)	2.961*** (0.560)	2.720*** (0.620)	2.861*** (1.223)	2.911*** (1.432)	-1.590*** (0.396)	-0.433 (0.433)	-0.556 (0.378)
Capital intensity			-0.390 (0.399)	-0.435** (0.550)	-0.390 (0.550)	-0.390 (0.550)	-0.390 (0.550)	-0.390 (0.550)	-0.390 (0.550)
Lerner index			-0.0247 (0.173)	-0.0247 (0.173)	-0.0247 (0.173)	-0.0247 (0.173)	-0.0247 (0.173)	-0.0247 (0.173)	-0.0247 (0.173)
Logfixed assets in starting year			-0.0273 (0.284)	-0.0273 (0.284)	-0.0273 (0.284)	-0.0273 (0.284)	-0.0273 (0.284)	-0.0273 (0.284)	-0.0273 (0.284)
Export intensity			-0.680*** (0.172)	-0.680*** (0.172)	-0.680*** (0.172)	-0.680*** (0.172)	-0.680*** (0.172)	-0.680*** (0.172)	-0.680*** (0.172)
adj. R^2	0.183	0.301	0.260	0.231	0.045	0.097	0.164	0.176	-0.002
# Obs.	79	79	79	79	79	79	79	79	79

Notes: The table examines correlations between standardized benchmark distortion centrality $\xi^{10\%}$ and measures of government interventions. The benchmark distortion centrality is constructed by assuming imperfections $\gamma_i = 0.1$ for all i . Industry averages of firm-level variables are computed after dropping outliers at 1%. Standard errors are in parentheses. *** and ** indicate significance at the 1% and 5% level, respectively.

TABLE XII
SECTORS WITH HIGH DISTORTION CENTRALITY HAVE LARGER SOE PRESENCE

	All SOEs in 2007		SOEs established after year T			
			$T = 2000$	$T = 2001$	$T = 2002$	$T = 2003$
	(1)	(2)	(3)	(4)	(5)	(6)
$\xi^{10\%}$	7.577** (2.963)	7.808** (2.834)	2.960*** (1.059)	2.549*** (0.886)	2.123*** (0.725)	1.545** (0.619)
Capital intensity			0.914 (2.535)	0.774 (0.947)	0.715 (0.792)	0.602 (0.649)
Lerner index			-4.622** (2.193)	-2.191*** (0.850)	-1.997*** (0.685)	-1.611*** (0.561)
Logfixed assets in starting year			6.974*** (2.500)	2.042** (0.968)	1.832** (0.809)	1.245* (0.663)
Export intensity			-5.660** (2.218)	-2.013** (0.829)	-1.810** (0.693)	-1.484** (0.568)
adj. R^2	0.066	0.290	0.269	0.284	0.276	0.220
# Obs.	79	79	79	79	79	79

Notes: Outcome variable: SOE's Share of Sectoral Value Added in 2007. The table examines correlations between standardized benchmark distortion centrality $\xi^{10\%}$ and SOE's share of sectoral value added. SOEs are identified following [Heck and Song \(2015\)](#). The benchmark distortion centrality is constructed by assuming imperfections $\gamma_i = 0.1$ for all i . Standard errors are in parentheses. *** and ** indicate significance at the 1% and 5% level, respectively.

- SOE-related interventions also correlate positively with ξ ;
- total implied output gains are economically meaningful (order of a few percent for the benchmark, and larger under some estimated-wedge specifications).

8.7 Table XIV: counterfactual targeting in China

Same logic as the Korea counterfactuals:

- targeting based on size (Domar weights) or consumption shares performs poorly;
- targeting based on export intensity or intermediate intensity performs better;
- the “optimal assignment” that directly targets ξ would yield substantially larger gains than the actual pattern of interventions.

TABLE XIII
EVALUATING SECTORAL INTERVENTIONS IN MODERN-DAY CHINA

Distortion centrality specification	$ad(\xi)$	Aggregate gains ($\Delta \ln Y$) by intervention (in percentage points)			
		Subsidized credit	Tax incentive	SOEs	Total
Benchmark ($\xi^{10\%}$)	0.22	1.69	0.64	1.27	3.60
De Loecker and Warzynski	0.42	3.07	1.19	2.39	6.65
Foreign firms as controls	0.25	1.69	0.67	1.16	3.51
Rejman and Zingales	0.11	1.01	0.36	0.65	2.02
Sectoral profit share	0.17	1.20	0.47	0.95	2.62

TABLE XIV
POLICY COUNTERFACTUALS FOR MODERN-DAY CHINA

Specification for ξ	Total output gains across all interventions (in percentage points)				Output gains relative to real-world interventions (correlation between ξ and the counterfactual policy target)			
	$\xi^{10\%}$	B1	B2	B3	$\xi^{10\%}$	B1	B2	B3
Real-world interventions	3.60	6.65	3.51	2.02	2.62	100%	100%	100%
Counterfactual policy target								
CF1 Domar weight γ	-1.42	-2.67	-1.18	-0.83	-1.14	-39.0%	-38.6%	-41.3%
CF2 Consumption share β	-2.56	-4.62	-2.43	-1.44	-1.90	-71.1%	-69.6%	-71.3%
CF3 Export intensity	1.13	1.98	0.99	0.79	0.80	31.4%	29.9%	38.9%
CF4 Sectoral value-added	-1.30	-2.41	-1.11	-0.75	-0.95	-36.2%	-36.3%	-37.5%
CF5 Intern. exp. share	1.34	2.39	1.11	0.83	0.87	37.2%	35.9%	41.2%
CF6 Optimal assignment	5.33	10.18	5.85	2.97	3.97	148.1%	153.4%	166.8%

9 Conclusion

9.1 Summary

Liu (2019) develops a theory of **industrial policy in a multi-sector production network** in which sectors are linked by input-output relationships and markets are distorted by wedges (e.g., taxes, credit frictions, markups, policy distortions). The central contribution is to show that, in this second-best environment, the welfare impact of subsidizing a sector is **not** determined solely by how distorted the sector is “locally,” but by how the sector’s distortions **propagate through the entire network**. This propagation is summarized by a single sufficient statistic: **distortion centrality**.

Formally, the paper defines distortion centrality as

$$\xi_i \equiv \frac{\mu_i}{\gamma_i},$$

where μ_i is *sectoral influence* (a local measure of how much aggregate output responds to a marginal relaxation/expansion of sector i) and γ_i is the *Domar weight* (a measure of the sector’s equilibrium size/cost of scaling it up). Thus ξ_i is the “**bang-per-buck**” of subsidizing sector i relative to its size: it is high when the sector is *too small* relative to its network importance.

The paper then provides a transparent policy-evaluation formula. Let s_i denote policy spending per value added in sector i . A first-order approximation around the decentralized equilibrium implies

$$\Delta \ln Y \approx \text{Cov}(\xi, s), \quad (36)$$

where the covariance is taken across sectors using value-added shares as weights. Because $\mathbb{E}[\xi] = 1$ in the decentralized economy, uniform subsidies (s_i constant across sectors) generate *zero* first-order aggregate effect; aggregate gains require **targeting**—i.e., loading spending on high- ξ sectors and (locally) taxing low- ξ sectors.

Finally, the paper links the theory to empirics. Because misallocative effects accumulate through **backward demand linkages**, upstream sectors tend to have high ξ and become “sinks” for distortions. In economies with **hierarchical** production networks (empirically relevant for South Korea and China), high- ξ sectors tend to coincide with upstream heavy/chemical industries, and this conclusion is comparatively robust to uncertainty about the exact micro wedges.

9.2 Main conclusions (the paper’s key takeaways)

The conclusion section emphasizes four messages:

1. **Start with high distortion-centrality sectors.** Well-meaning interventions need not target the *most locally distorted* sectors, and subsidizing relatively undistorted sectors need not worsen misallocation. This echoes the logic of the second best, but ξ provides a succinct quantitative summary of network misallocation.
2. **Policy evaluation reduces to a covariance.** Proposition 2 implies that, to first order, aggregate gains are increasing in the covariance between policy spending and distortion centrality (Eq. (36)). This provides a practical way to evaluate and compare counterfactual industrial policies even in general equilibrium.
3. **Upstream sectors tend to have high ξ .** Because distortions compound through backward linkages, upstream industries tend to have larger distortion centrality. In South Korea and China, hierarchical network structure makes heavy/chemical sectors systematically higher- ξ than light manufacturing.
4. **Distortion centrality predicts actual interventions and implies sizable aggregate gains.** In the empirical applications, sectoral interventions (credit availability, tax incentives, and SOE-directed funds) align with ξ to a meaningful extent. Quantitatively, the estimated interventions increase aggregate consumption in China by about **3.5–6.7%**, and South Korea’s HCI-style sectoral spending generates positive net effects.

9.3 Economic intuition: why distortion centrality is the “right” targeting object

The intuition is a network multiplier logic:

1. **A local distortion shrinks a sector and depresses upstream demand.** If sector i is inefficiently “too small” because of a wedge, it buys fewer inputs from its suppliers. Those suppliers then become too small, and so on upstream.
2. **Therefore distortions compound as we move upstream.** Backward linkages create a cascade: a downstream wedge generates an upstream sequence of demand shortfalls. Upstream sectors inherit the downstream sector’s imperfection and the imperfections of all other downstream users.
3. **A subsidy relaxes a wedge and triggers a cascade in the opposite direction.** Subsidizing sector i expands its activity and raises demand for its suppliers’ inputs, raising suppliers’ scale and output, and propagating upstream through the network.
4. **Benefit versus cost: influence vs. Domar weight.** The *benefit* of subsidizing a sector is proportional to its *influence* μ_i (how much aggregate output responds when the sector expands). The *fiscal cost* is proportional to its equilibrium size γ_i (the amount of spending needed per unit of aggregate scaling). Their ratio $\xi_i = \mu_i/\gamma_i$ is exactly “aggregate gain per dollar” at the margin.

9.4 Caveats (limits explicitly acknowledged by the paper)

The paper emphasizes that its sufficient-statistic results are **local/marginal**:

- it does not characterize when subsidies become “too large”,
- it abstracts from political economy and implementation constraints,
- it treats wedges as policy-invariant even though policy may change wedges,
- it focuses on static reallocation effects and omits dynamic considerations (e.g., learning, innovation).

These limitations naturally motivate future work that combines ξ -style targeting with dynamic mechanisms and endogenous policy distortions.

9.5 Takeaway

In production networks, the welfare impact of industrial policy is governed by how distortions propagate through input–output linkages, and distortion centrality ξ summarizes the marginal “bang-per-buck” of subsidizing a sector; consequently, aggregate gains are (locally) captured by the covariance between policy spending and ξ , and upstream sectors tend to be the highest-return targets in hierarchical networks.